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Abstract

The Buchwald–Hartwig cross-coupling is a cornerstone of modern pharmaceutical synthesis, yet predictive modeling of its outcomes remains limited by data quality and chemical space coverage. Industry electronic laboratory notebooks (ELNs) contain heterogeneous, noisy records, while open-source high-throughput experimentation (HTE) datasets are fragmented and narrow in scope. As a result, models often fail when applied to novel substrate and condition combinations. Here we introduce a unified framework that systematically standardizes and integrates diverse reaction data into a high-quality, unique-structure-per-entity dataset, coupled with active learning to strategically expand chemical space. By merging published Buchwald–Hartwig HTE data with new experimental results, we achieve models that generalize across substrates and conditions, delivering substantially improved out-of-distribution predictions relative to previous approaches. Crucially, model-guided reagent and condition recommendations were validated experimentally, confirming the framework's utility for exploring unexplored reactivity. This work establishes a blueprint for robust machine learning in synthetic chemistry, with the potential to accelerate pharmaceutical discovery by enabling more reliable and scalable prediction of reaction outcomes.

Keywords

high-throughput experimentation, machine learning, Buchwald-Hartwig coupling, synthesis prediction

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Abstract: The Buchwald–Hartwig cross-coupling is a cornerstone of modern pharmaceutical synthesis, yet predictive modeling of its outcomes remains limited by data quality and chemical space coverage. Industry electronic laboratory notebooks (ELNs) contain heterogeneous, noisy records, while open-source high-throughput experimentation (HTE) datasets are fragmented and narrow in scope. As a result, models often fail when applied to novel substrate and condition combinations. Here we introduce a unified framework that systematically standardizes and integrates diverse reaction data into a high-quality, unique-structure-per-entity dataset, coupled with active learning to strategically expand chemical space. By merging published Buchwald–Hartwig HTE data with new experimental results, we achieve models that generalize across substrates and conditions, delivering substantially improved out-of-distribution predictions relative to previous approaches. Crucially, model-guided reagent and condition recommendations were validated experimentally, confirming the framework's utility for exploring unexplored reactivity. This work establishes a blueprint for robust machine learning in synthetic chemistry, with the potential to accelerate pharmaceutical discovery by enabling more reliable and scalable prediction of reaction outcomes.

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Introduction

The Buchwald-Hartwig (BH) cross-coupling reaction[1, 2] has become an indispensable tool in modern synthetic chemistry[3,4], particularly in the pharmaceutical industry, where it is used to synthesize structurally diverse (hetero)aryl amines compounds[5]. The ability to reliably forecast the outcomes of BH reactions is thus invaluable, with Machine-Learning (ML) methods offering a pathway to streamline and accelerate lead discovery efforts[6,7]. A recent example introduced a tool[8] to estimate substrate-adaptive conditions for palladium-catalyzed C–N couplings, trained on a diverse experimental dataset of reactant pairings and systematically refined through active learning. Beyond predicting reaction outcomes[9,10,11,12], the integration of reagent dictionaries[13] enables yield prediction models to suggest the most suitable catalysts, bases, solvents and other conditions to achieve desired transformations. Retrosynthesis planning[14,15,16], which identifies viable synthetic routes to target molecules, can further benefit from accurate reaction-level predictors, as robust models of cross-coupling chemistry directly expand the reliability of multi-step route design. However, the inherent complexity and sensitivity to conditions in these reactions pose significant challenges for traditional predictive models[17,18,19]. Addressing these complexities requires not only vast amounts of data but also a degree of consistency and precision in data generation that is often difficult to achieve, in part, due to the heterogeneity of the data sources.

Mining data from electronic lab notebooks (ELN) is one possible avenue towards building predictive synthesis models, but the performance of these models is capped due to data quality. ELN and literature scraped datasets[20] can vary widely in reaction setup, purification processes, and yield reporting. Additionally, such datasets have heterogeneous, and sometimes contradictory representations of the same chemical entities, do not capture all reaction components in a tabular format and, in the case of literature, lack negative examples.

While better algorithms are still required to tackle this industry-wide challenge [21] , HTE has emerged as a powerful method for generating reaction data with high consistency and quality through controlled and automated processes. Recently, a large-scale release of over 39,000 previously proprietary HTE reactions has further enriched the chemical data landscape[22], enabling systematic exploration which unveiled hidden statistical relationships between reaction components and uncovered dataset biases.

Similarly, parallel library synthesis has been instrumental in drug discovery, enabling the efficient production of structurally diverse analogues through robust synthetic methodologies[23]. Dombrowski et al. [24] analyzed over 5,000 parallel libraries across 14 years, showcasing the evolution and proven success of these approaches. Notably, Buchwald–

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Hartwig cross-coupling reactions were identified as one of the most challenging, with among the lowest success rates. Nonetheless, HTE platforms have demonstrated notable success in optimizing BH cross-coupling reactions, as reported by Dreher et al.[25], offering robust tools for synthesizing complex molecules on micro- and nanomolar scales and under automation-friendly conditions.

Collectively, these advancements established a foundation for reliable and standardized data generation, essential for navigating the inherent complexities of predictive BH reaction modeling. Moreover, HTE allows for the quick accumulation of a vast number of data points tailored to specific needs, encompassing a wide range of substrates and diverse reaction conditions. The ability to efficiently explore diverse reaction environments and generate comprehensive data makes HTE an ideal source of data for ML models aimed at estimating reaction outcomes, particularly when robust out-of-distribution (OOD) performance is required.

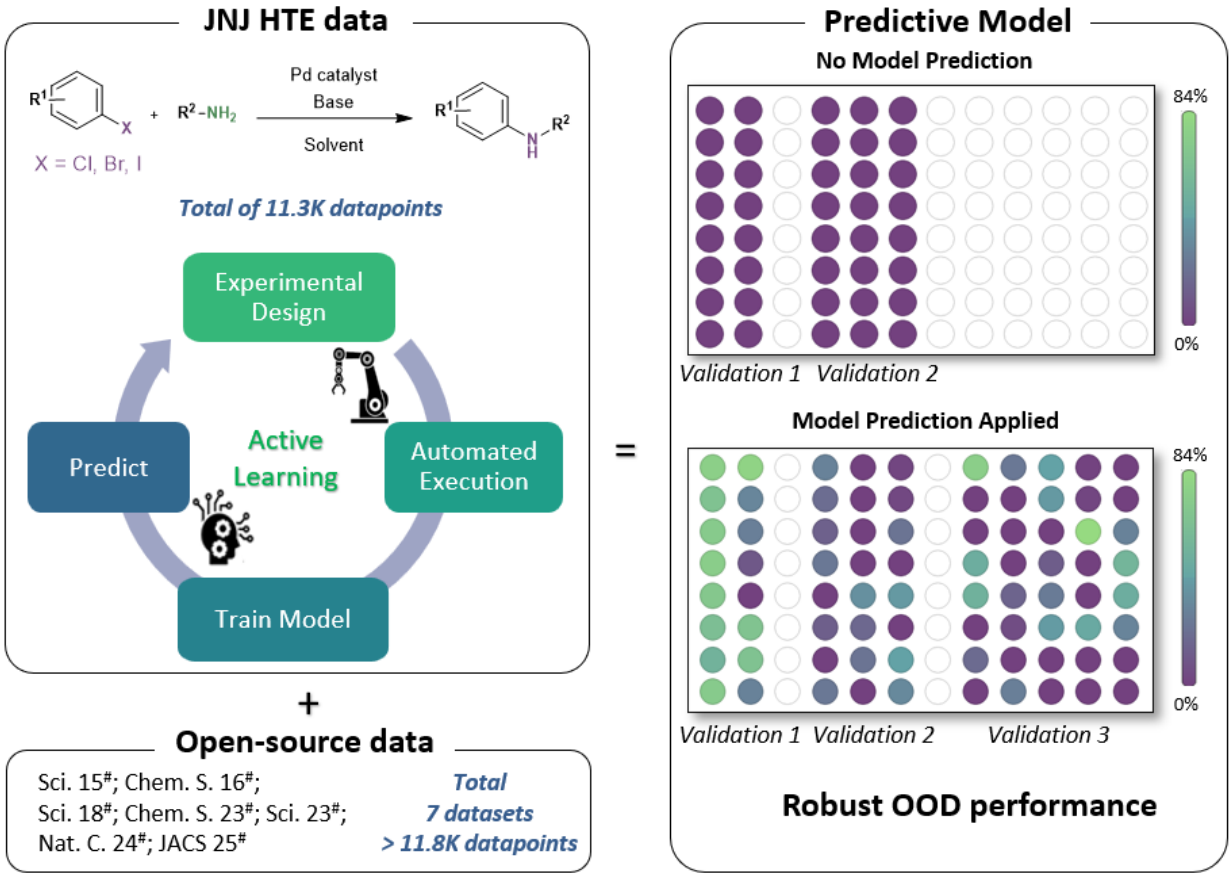

Figure 1 – Project overview: the combination of internally generated datasets and open source HTE data leads to a model with robust predictive performance in three scenarios: a. identifies and corrects mislabeled data, b. rescues previously unsuccessful reactant pairs and c. enables challenging, out-of-distribution substrate couplings.

ML models excel at identifying complex patterns within large datasets[26], and when trained on high-quality data, such as HTE data, they can more accurately predict reaction outcomes,

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even for novel substrates. This is the reason why in recent years there has been an increase in studies combining ML and HTE[8,27,28,29]. The integration of HTE and ML therefore has the potential to transform reaction prediction by delivering models capable of OOD generalization—where predictions remain reliable despite differences between training and test data. This synergy is particularly impactful for applications in synthetic and medicinal chemistry, where the ability to forecast reaction outcomes in new, untested scenarios can significantly improve the time and resource efficiency required for chemical exploration and development.

The goal of this study is to develop the first predictive model with wide OOD capabilities for BH reactions by integrating a newly generated HTE dataset with available open-source HTE data. To this end, we employed a data-driven active learning[30] approach that iteratively refined the model's accuracy and generalization capacity across diverse chemical spaces. By standardizing and unifying disparate datasets into a single, high-quality format, we have created a comprehensive dataset that supports a model capable of reliable predictions across a meaningful area of the BH reaction landscape. This work not only advances the state of predictive chemistry for a critical reaction type but also demonstrates the potential of HTE and ML integration as a general framework for achieving robust, adaptable reaction prediction models and reagent recommender systems.

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Results

In this project, we strategically leverage automated workflows using 96-well and 384-well plates to execute our HTE designs, with product quantification performed using UPLC[31]-MS-CAD (Charged Aerosol Detector)[32](Figure 2). We employed an iterative approach for data generation, purposefully selecting experimental conditions across three iterations. The first iteration focused on establishing key variables and increasing diversity in reaction conditions by exploring combinations of catalysts, bases, and solvents. For iterations 2 and 3, we employed model-guided experimental design, targeting broader coverage of the substrate chemical space. Reactions with high model uncertainty were prioritized, enabling continuous refinement of the predictive model's accuracy. This systematic approach significantly enhanced data diversity and quality, resulting in the largest and most comprehensive BH HTE dataset in literature to date with 11.3k high-quality reactions. The production of this data followed a quantitative methodology consisting of two large phases. The first phase aimed to produce a starting baseline ML model, which was used in a second phase to perform experimental design in an active learning formulation. Additionally, we curated and standardized all other published BH HTE datasets into a unified format, creating an extensive dataset with 24.5k reactions.

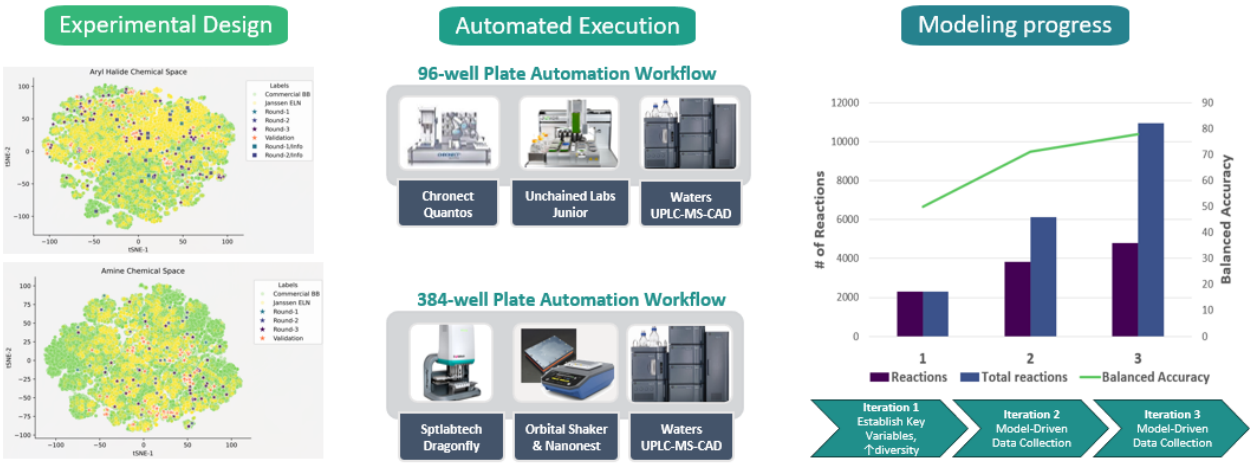

Figure 2 – Experimental design and execution of HTE campaigns towards the build of a generalizable BH model; the bar graph showcases how an active learning approach leads to high model performance as shown by the balanced accuracy metric.

The strategy for the first phase was centered around four factors: (1) informer set, (2) structure diversity criteria, (3) space density criteria and (4) undesired substructure filtering. (Supporting Information - Section 1)

The second phase embraced an active learning methodology utilizing (1) BERT-Enriched-Embeddings[13] (BEE) an extension of a language model for yield prediction[17], which was trained on the data from the previous phase, (2) exploration of a virtual BH space of 64M

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reactions, (3) Ranked Batch-Mode Active Learning[33] (RBMAL) function using model uncertainty and Tanimoto distance to previous selections and (4) diversity & density. (Supporting Information - Section 2)

We address model bias in benchmarking by training three ML models and one AI SMILES based model, and we address data bias[34] by testing on OOD data. OOD was created by clustering all available reactions with differential reaction fingerprint[35] (DRFP) and Kmeans[36] and only testing reactions absent in the clusters which contained reactions used in the training. Finally, we addressed metric bias[37] by considering a total of 10 performance metrics and identifying 4 which can summarize the capabilities of interest (uncertainty estimation, good precision and serviceable recall in class imbalance OOD scenarios) of the different models.

The data produced in this study (labeled JnJ24 in Figure 3) not only surpasses previous studies in overall size but also in diversity of reagent combinations, diversity of substrate pairs and number of unique substrates, see Figure 3 a). Produced data is represented as "JnJ 24" and the combination of this data with the remaining open-source datasets is represented in "All w JnJ24". Relevant research regarding HTE data production in Pd catalyzed C-N couplings (4204 reactions) for ML modelling was released during the time of the writing of this manuscript[38] by a partnership between Merck and MIT; although not featured in the analysis in Figure 3, the data is included in the global HTE open-source dataset released in this project.

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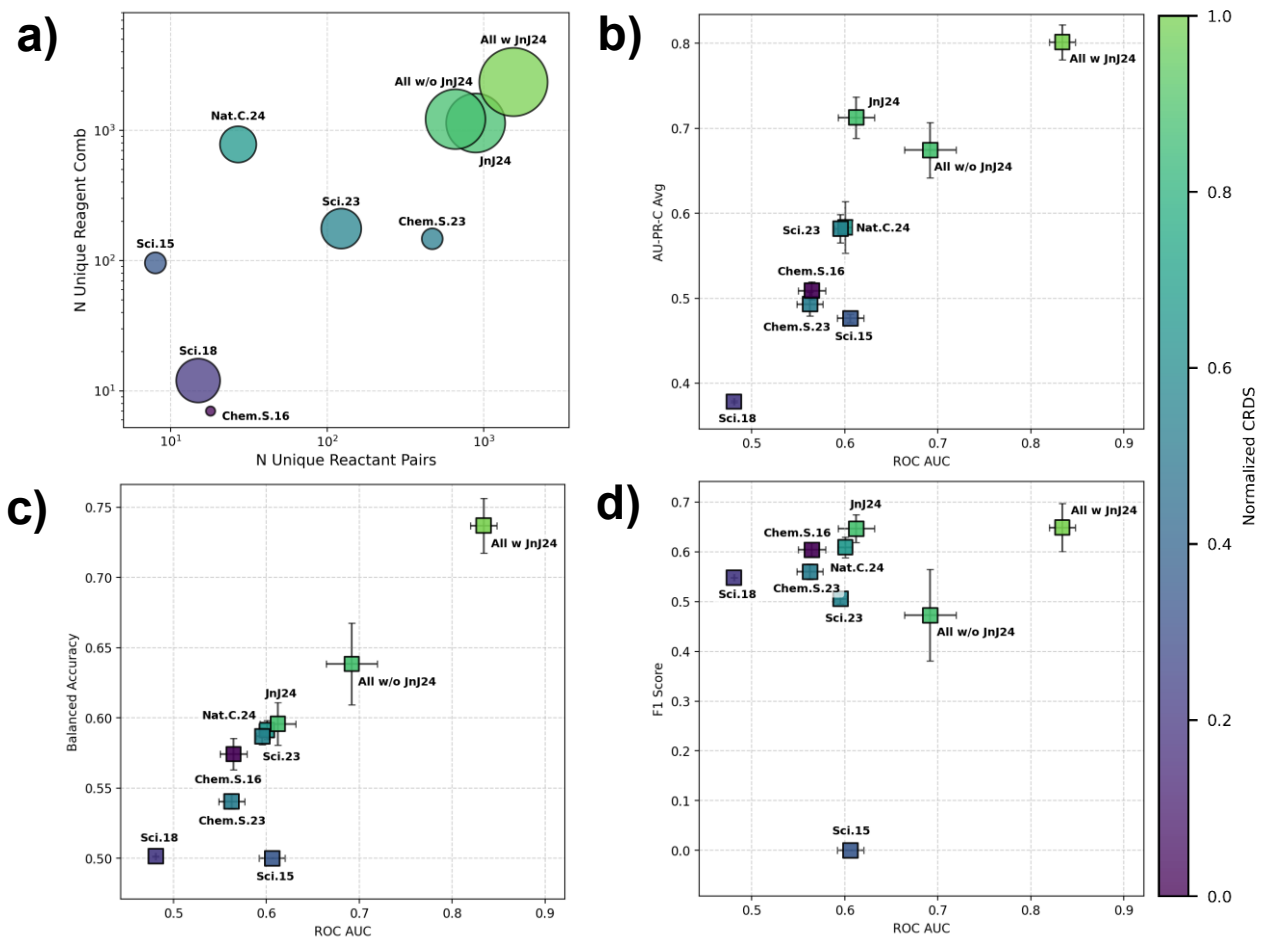

Figure 3 – a) Visual representation of unique substrate pairs, unique reagent combinations and size for each of the 7 open-source BH HTE datasets used in project, colored by CRDS (data-centric score with strong correlation to performance). b) c) d) performance across four critical metrics, for the best ML model for each given dataset when tested on OOD clusters built from all remaining open-source data.

We introduce a new data centric metric Compound-Reaction Diversity Score (CRDS) which aims to estimate the ability of potential models trained on a given dataset to perform in OOD scenarios. This formula combines the number of unique reactant pairs, unique reagent combinations and size of the dataset. Additionally, each component has adjustable thresholds to induce diminishing returns and varying weights. When applying this calculation, we found a 96.7 Spearman correlation between CRDS and ROC AUC when in OOD testing, indicating a high degree of correlation between data diversity-quantity and OOD performance, see Table 1. More details about CRDS formula and calculation can be found in the Supporting Information (See SI, Section 2.2).

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	Pearson				Spearman			
	ROC AUC	Bal. Acc.	F1 Score	AU-PR-C	ROC AUC	Bal. Acc.	F1 Score	AU-PR-C
CRDS	**0.897**	0.814	0.409	**0.889**	**0.967**	**0.900**	0.517	**0.967**
Simplified CRDS	0.884	**0.820**	**0.449**	0.865	0.933	0.833	**0.600**	**0.967**
Dataset Size	0.588	0.614	0.193	0.667	0.650	0.700	0.117	0.617

Table 1 - Pearson and Spearman correlation between CRDS, simplified CRDS and Size of the Train Dataset against ML model performance on synthesis success prediction of reactions in an OOD Cluster.

Furthermore, JnJ 24 surpasses the number of unique substrate pairs even when compared to the combined effort of the 6 previous publications represented in All w/o JnJ24. In Figure 3 b), c) and d), it's observed that the collection of all datasets while excluding the JnJ24 data, surpasses performance of models trained on previous individual works in some crucial metrics such as ROC AUC, Balanced Accuracy and AU-PR-C, but the best model trained on this data is then significantly surpassed in all metrics when combined with JnJ24 into a single dataset.

We also demonstrate how a binary reactivity classification model can be used as a reagent recommender and HTE plate designer (recommending both substrates and reagents), when combining it with a reagent dictionary and uncertainty estimation.

Before utilizing these types of models in the lab or executing experimental validation, it's important to assess whether the models are uncertainty calibrated, as uncertainty will be used to rank reagent and substrate pair combinations. We tackle this by binning the predictions of OOD reactions into 5 confidence bins. The confidence value is inversely derived from uncertainty and normalized between 0.5 and 1 as described in[13] If the model is perfectly calibrated then performance would perfectly correlate with confidence, following the gray dashed line in Figure 4.

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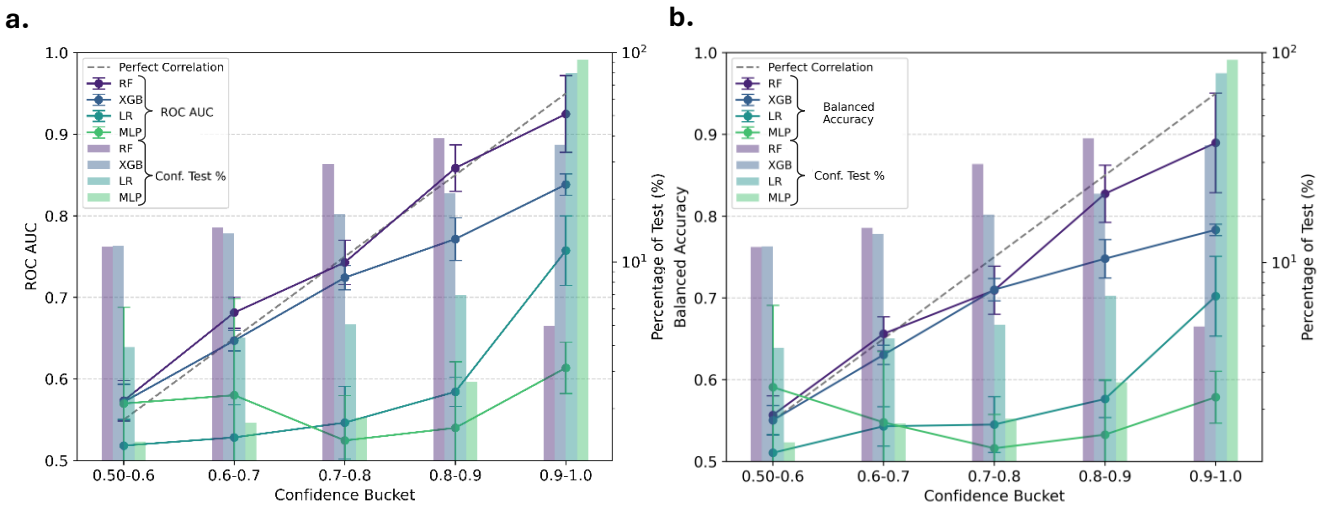

Figure 4 – Data was clustered 20 times to create train-test splits with in-domain and out-of-domain reactions. Four different ML models were train-tested and using model confidence predictions were grouped into 5 buckets. A) and B) show how ROC AUC and Balanced Accuracy respectively vary for each model and confidence. Random Forest stands out as having good correlation between confidence bucket and model performance.

Through the use of the confidence metrics it is possible to rank reagent combinations for any given substrate pair, by their likelihood of success. Unexpectedly, we discovered that model performance on out-of-distribution predictions was not primarily determined by dataset size, but rather by a previously unrecognized relationship between data diversity and prediction accuracy, as shown in table 1a. Through development of a new metric, the Compound-Reaction Diversity Score (CRDS), we revealed a striking 96.7 Spearman correlation between dataset diversity and out-of-distribution performance. This finding challenges the conventional wisdom that larger datasets necessarily lead to better generalization, instead showing that strategic diversity in reaction space coverage is the key driver of robust predictions, see Supporting Information - Section 3 for formulation. Furthermore, we found that a relatively simple Random Forest[39] model, when trained on our properly curated dataset, achieved unprecedented >90% precision on high-confidence out-of-distribution predictions, demonstrating that model architecture sophistication may be less critical than previously thought, at least for achieving reliable chemical prediction in a single reaction type with less than 25k samples.

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Experimental Validation Results

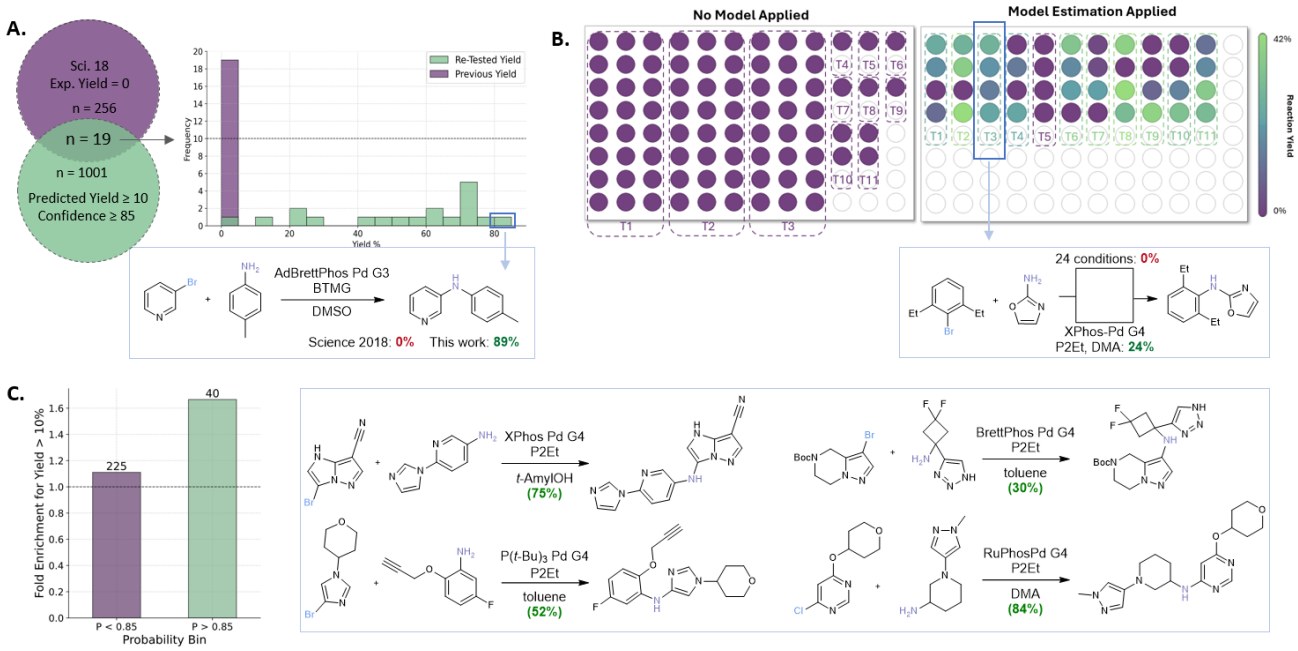

Figure 5 - a) The model identified 19 reactions previously reported as unsuccessful but predicted to have yields above 10% with high confidence (>85%). Experimental retesting confirmed successful product formation in 18 of these reactions. b) Model-guided rescue of historically unsuccessful substrates. Using model recommendations, 11 previously unsuccessful aryl–amine pairs were tested across the four highest-ranked reaction conditions achieving viable products for 10 of the 11 substrate pairs. c) Prospective validation of the model across a novel chemical reaction space. Experimental results present fold enrichment of reactions selected from a vast chemical domain (~2.46×10^{11} reactions) predicted by the model with moderate confidence (<0.85) and high confidence (>0.85) versus a baseline set.

One critical observation from our analysis was the potential impact of mislabeled reactions within legacy datasets, which could significantly hinder predictive model accuracy despite advancements in dataset diversity. To address this, the first experiment focused on the identification and correction of such mislabeled reactions (Figure 5a). To validate our model's ability to rectify data quality issues, we revisited published HTE data from a 2018 *Science* publication[27], concentrating on reactions previously reported with 0% yield. The model flagged 19 of these 256 negative outcomes as potentially mislabeled, predicting yields above 10% with confidence exceeding 85%. Upon experimental re-validation under original conditions, 18 of these flagged reactions indeed yielded products above the 10% threshold (95% accuracy). This result uncovered critical mislabeling within legacy datasets, demonstrating the model's capability to enhance dataset integrity and improve reliability in downstream predictions.

The second experiment was aimed at rescuing historically unsuccessful substrates (Figure 5b). In this scenario, we tested the practical applicability of our model by using it as a reagent recommender[13] to identify successful reaction conditions for 11 historically unsuccessful

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aryl halide–amine substrate combinations gathered from HTE literature reactions as well as data produced during this project. Previously, three substrate pairs had each failed across 24 reaction conditions, two pairs had each failed across two reaction conditions, and the remaining six pairs had each failed in one reaction condition. We then selected catalysts, bases and solvents that were frequently covered in the dataset (minimum 400 samples), leading to a total of 1,092 reagent condition combinations. Leveraging the model, we systematically ranked all these combinations according to predicted confidence. For each substrate pair, only the top four recommendations were experimentally tested. Out of these 44 reactions, 27 succeeded (yield >10%), efficiently providing viable synthetic routes for 10 of the 11 substrates (91%). This illustrates the model's ability to efficiently identify effective reaction conditions, significantly reducing the experimental burden and conserving laboratory resources.

The third experiment, Figure 5 c), was a prospective validation across novel chemical space. We assessed the model's predictive generalizability in a vast, unexplored chemical domain sampling from a virtual space consisting of 13 catalysts, 12 bases, 7 solvents, 12,252 aryl halides, and 18,364 amines, representing approximately 2.46×10^{11} possible Buchwald-Hartwig reactions. We specifically selected entirely novel aryl and amine combinations (absent from any prior training data) to rigorously test the model's ability to identify successful reactions. From this space, we experimentally evaluated three groups of reactions:

- A baseline group of 4,800 reactions, selected via diversity, space density, pharmacological relevance, and model uncertainty, which yielded around 19% success rate.
- Reactions predicted with moderate confidence (<0.85, for reference 0.5 indicates complete uncertainty) achieved a success rate of 21%, a fold-enrichment of 1.11 over the baseline, see Figure 5 c).
- Reactions predicted with high confidence (>0.85) achieved a success rate of 33%, representing a fold enrichment of 1.64 over the baseline, see Figure 5 c).

Critically, these results highlight the model's exceptional capacity to distinguish reactions where modeling can enrich from those out-of-domain, even when dealing with entirely novel chemical combinations drawn from an extraordinarily expansive chemical space.

Collectively, these experiments validate that our unified, actively curated machine learning model substantially enhances Buchwald-Hartwig reaction predictions by (i) correcting mislabeled historical data, (ii) effectively rescuing historically unsuccessful reaction pairs through minimal additional experimentation, and (iii) confidently identifying viable reactions in previously unexplored chemical space. This approach offers a powerful, transferable

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framework for accelerating the discovery and optimization of pharmaceutically relevant chemical transformations and the growing high quality curated dataset enables further modeling research.

Conclusion and Outlook

We have developed a unified, actively curated machine learning framework capable of robust out-of-distribution prediction for Buchwald–Hartwig cross-coupling reactions. By systematically standardizing heterogeneous datasets, strategically expanding reaction space through active learning, and validating predictions experimentally, we demonstrate that data diversity, rather than dataset size or model complexity alone, is the key driver of generalizable performance. This insight not only advances predictive chemistry for a pivotal reaction class but also establishes principles broadly applicable to other transformations.

An opportunity for future work remains the harmonization of noisy, heterogeneous reaction data (eg, industry ELNs), inclusion of conditions in a global setting and scaling predictive frameworks across reaction types. Better algorithms are still required to tackle these industry-wide challenges; potentially in the future the use of knowledgeable chemistry Artificial Intelligence (AI) agents[40,41] will make possible a harmonization of the free-text protocol and tabular data, while noise detection techniques will be able to remove the most "obvious" noisy samples from the training dataset[42,43,44]. Looking ahead, the integration of additional high-throughput experimentation campaigns, richer reaction context representations can further enhance predictive power.

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